New lower bounds for line-free sets in $\mathbb{F}_7^3$ and $\mathbb{F}_{11}^3$

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Abstract

A subset of $\mathbb{F}_p^n$ is *line-free* if it contains no affine line, equivalently no p-term arithmetic progression with nonzero common difference; $r_p(\mathbb{F}_p^n)$ denotes the largest size of such a set. We exhibit an explicit line-free set of 227 points in $\mathbb{F}_7^3$ and an explicit line-free set of 1010 points in $\mathbb{F}_{11}^3$, so that $r_7(\mathbb{F}_7^3) \geq 227$ and $r_{11}(\mathbb{F}_{11}^3) \geq 1010$; the previously best known lower bounds were 225 and 1005. By the product construction, $r_7(\mathbb{F}_7^{3k}) \geq 227^k$ for every $k \geq 0$, and hence $\lim_{n\to\infty} r_7(\mathbb{F}_7^n)^{1/n} \geq 227^{1/3} \approx 6.1002$; this improves the three-dimensional product bound $225^{1/3} \approx 6.0822$ of Elsholtz et al., but not the best known bound $7^{20/21} \approx 6.3805$, which follows from the line-evasive sets of Führer and Taranchuk (2024). The sets were found by tabu search with iterated local search. Both lower bounds and the statement $r_7(\mathbb{F}_7^{3k}) \geq 227^k$ for all k are certified in Lean 4 (core library only) by kernel-checked computation and a structural proof; the only step not formalised is the passage to the limit (Fekete's lemma).

Contents

1 Introduction **2**

2 Definitions and results **3**

3 The two sets **4**
3.1 Encoding and verification 4
3.2 Statistics 4

4 How the sets were found **5**

5 Formal verification **5**

6 Limitations and open questions **7**

A The 227-point line-free set in $\mathbb{F}_7^3$ **7**

B The 1010-point line-free set in $\mathbb{F}_{11}^3$ **7**

1 Introduction

For a prime p and integers $n \geq 1$, $3 \leq k \leq p$, let $r_k(\mathbb{F}_p^n)$ denote the maximum size of a subset of $\mathbb{F}_p^n$ containing no k-term arithmetic progression $a, a+d, \ldots, a+(k-1)d$ with $d \neq 0$. The case $k = p = 3$ is the cap set problem. For $k = p$ a nontrivial p-term progression is exactly an affine line $\{a + td : t \in \mathbb{F}_p\}$, so $r_p(\mathbb{F}_p^n)$ is the largest size of a *line-free* set. Equivalently, $p^n - r_p(\mathbb{F}_p^n)$ is the smallest size of a set of points of the affine space $\mathrm{AG}(n,p)$ meeting every line (for $n = 3$: a 2-blocking set).

Line-free sets were studied systematically by Elsholtz, Führer, Füredi, Kovács, Pach, Simon and Velich [1]. The hypercube $\{0, 1, \ldots, p-2\}^n$ is line-free, so $r_p(\mathbb{F}_p^n) \geq (p-1)^n$, with equality for $n \leq 2$ (Jamison [9], Brouwer and Schrijver [10], as quoted in [1]). In dimension three, [1] proves

$$(p-1)^3 + p - 2\sqrt{p} \;\leq\; r_p(\mathbb{F}_p^3) \;\leq\; p^3 - 2p^2 - (\sqrt{2}-1)p + 2 \qquad (p \geq 5),$$

a further general bound $r_p(\mathbb{F}_p^n) \geq (p-1)^n + \frac{n-2}{2}(p-1)(p-2)^{n-3}$ (their Theorem 1.7, which for $n = 3$ gives $(p-1)^3 + \frac{p-1}{2}$), the special bound $r_7(\mathbb{F}_7^3) \geq 225$ (their Theorem 1.2 and the remark following its proof), a 70-point line-free set in $\mathbb{F}_5^3$ found by a branch-and-cut computation, and the upper bounds $r_5(\mathbb{F}_5^3) < 74$, $r_7(\mathbb{F}_7^3) < 243$. Kovács [2] recently improved the lower bound to $p^3 - 3p^2 + \frac{1}{8}p^{3/2} - \frac{7}{16}p = (p-1)^3 + \frac{1}{8}p^{3/2} - O(p)$; his construction uses $\lfloor \sqrt{p}/4 \rfloor$ special layers and is void for $p < 16$, and the explicit bound exceeds $(p-1)^3$ only for $p \geq 757$, so it gives nothing for $p = 7, 11, 13$. [2] states that, to the author's knowledge, there had been no further improvements of the bounds of [1] from either side.

A further classical construction is the *cone* (S. Ball's lecture notes, as quoted in the MathOverflow discussion [4] opened by A. Bishnoi in January 2016): if B is a double blocking set of $\mathrm{PG}(2,p)$, i.e. a set meeting every line in at least two points, then the affine part of the union of the lines joining a fixed affine point to the points of B meets every affine line, so that $r_p(\mathbb{F}_p^3) \geq p^3 - 1 - (p-1)|B|$. With the sides of a triangle ($|B| = 3p$) this is exactly $(p-1)^3$. Double blocking sets of size $3p-1$ in $\mathrm{PG}(2,p)$ are known for $p = 13$ (Braun, Kohnert and Wassermann [6]; Csajbók and Héger [7]) and for $p = 19, 31, 37, 43$ [7]; none exists for $p \leq 8$ (Ball and Blokhuis [5]), and for $p = 11$ none with fewer than $3p = 33$ points is known ([7]; the lower bound is 31 [5]). For $p = 13$ the cone over the 38-point set of [7, §4.1] is therefore a line-free set of $13^3 - 1 - 12 \cdot 38 = 1740$ points (we verified this by computer; see `data/cone13_1740.py`), which is larger than the 1734 points of [1, Thm. 1.1 and 1.7]; Table 1 takes this into account. Zare's single-point improvement $(p-1)^3 + 1$ [4], mentioned in [1], and the constructions of affine 2-blocking sets of Bishnoi, D'haeseleer, Gijswijt and Potukuchi [8], which concern growing dimension, do not affect the small cases.

Since products of line-free sets are line-free (Theorem 2.4), $r_p(\mathbb{F}_p^n)$ is supermultiplicative in n, and the limit $\alpha_p = \lim_n r_p(\mathbb{F}_p^n)^{1/n}$ exists. Small-dimensional examples give lower bounds for α_p; [1] obtained $\alpha_5 \geq 70^{1/3} \approx 4.121$ and $\alpha_7 \geq 225^{1/3} \approx 6.082$, which were better than the bounds 4.090 and 6.066 (their Table 1) obtained from Frankl, Graham and Rödl's construction in dimension $2p$. These bounds have since been superseded: Theorem 1.1 of Führer and Taranchuk [3] (first version March 2024) partitions $\mathbb{F}_q^n$, $n = \binom{t+1}{2}$, into q sets each meeting every line in at most t points; for $t = p-1$ these are line-free sets of size p^{n-1} in $\mathbb{F}_p^n$, $n = p(p-1)/2$, whence $\alpha_p \geq p^{1-2/(p^2-p)}$, i.e. $\alpha_5 \geq 5^{9/10} \approx 4.2567$, $\alpha_7 \geq 7^{20/21} \approx 6.3805$ and $\alpha_{11} \geq 11^{54/55} \approx 10.5307$. Since $4.2567^3 > 73$, $6.3805^3 > 242$ and $10.5307^3 > 1085$ exceed the upper bounds for $r_p(\mathbb{F}_p^3)$, no three-dimensional set can reach these values, and the product bound of Theorem 2.6 below is not a record.

Table 1 summarises the state of the art for small p before this note.

p	lower bound	upper bound	source of the lower bound
5	70	73	computer search (branch and cut) [1]
7	225	242	algebraic construction, [1, Thm. 1.2]
11	1005	1085	$(p-1)^3+(p-1)/2$, [1, Thm. 1.1 and 1.7]
13	1740	1855	cone over a 38-point double blocking set of PG$(2,13)$ [6, 7]

Table 1: Bounds for $r_p(\mathbb{F}_p^3)$ before this note. Upper bounds: [1, Thm. 1.5, 1.6] for $p=5,7$ and the exact form of [1, Cor. 1.4] for $p=11,13$ (its simplified form $p^3-2p^2-(\sqrt{2}-1)p+2$ gives 1086 for $p=11$). The entry 1740 for $p=13$ combines known results (cone construction and [6, 7]) and is not stated explicitly in the literature we found.

What is new. We improve the lower bounds for $p=7$ and $p=11$ (Theorems 2.2 and 2.3) by two and five points respectively; the exact values remain open. The product bound for α_7 (Theorem 2.6) is included for completeness only, since it is weaker than the bound of [3]. To our knowledge, as of October 8, 2026, 227 and 1010 are the best known lower bounds for $r_7(\mathbb{F}_7^3)$ and $r_{11}(\mathbb{F}_{11}^3)$; we searched arXiv and the web for later work on line-free sets and on 2-blocking sets in $\mathrm{AG}(3,p)$ and found no improvement of the values in Table 1 (in particular none in [2, 3]; the cone construction gives 216 for $p=7$ and at most 1000 for $p=11$ with the known double blocking sets). The sets were found by computer search; they are irregular (the 227-point set has trivial stabiliser in $\mathrm{AGL}(3,7)$) and we do not have a conceptual description of them. The proofs consist of an exhaustive verification, which we also certify in Lean 4.

2 Definitions and results

Definition 2.1. Let p be a prime. A *line* of $\mathbb{F}_p^n$ is a set $\{a+td : t\in\mathbb{F}_p\}$ with $a,d\in\mathbb{F}_p^n$, $d\neq 0$. A set $S\subseteq\mathbb{F}_p^n$ is *line-free* if it contains no line. $r_p(\mathbb{F}_p^n)$ is the maximum size of a line-free subset of $\mathbb{F}_p^n$, and $\alpha_p=\lim_{n\to\infty} r_p(\mathbb{F}_p^n)^{1/n}$.

Theorem 2.2. *The set $S_7\subseteq\mathbb{F}_7^3$ of 227 points listed in Section A is line-free. Hence $r_7(\mathbb{F}_7^3)\geq 227$.*

Theorem 2.3. *The set $S_{11}\subseteq\mathbb{F}_{11}^3$ of 1010 points listed in Section B is line-free. Hence $r_{11}(\mathbb{F}_{11}^3)\geq 1010$.*

Lemma 2.4 (product construction)**.** *If $A\subseteq\mathbb{F}_p^m$ and $B\subseteq\mathbb{F}_p^k$ are line-free, then $A\times B\subseteq\mathbb{F}_p^{m+k}$ is line-free.*

Proof. Let $L=\{a+td\}$ be a line of $\mathbb{F}_p^{m+k}$ and write $a=(a_1,a_2)$, $d=(d_1,d_2)$. If $d_1\neq 0$, the projection $\{a_1+td_1\}$ is a line of $\mathbb{F}_p^m$, so some $a_1+t_0d_1\notin A$, and then $a+t_0d\notin A\times B$. If $d_1=0$ then $d_2\neq 0$ and the same argument applies to the second factor. □

Corollary 2.5. *For every $k\geq 0$, $r_7(\mathbb{F}_7^{3k})\geq 227^k$ and $r_{11}(\mathbb{F}_{11}^{3k})\geq 1010^k$.*

Proof. Apply Theorem 2.4 $k-1$ times to S_7 (resp. S_{11}); $|S^k|=|S|^k$. □

Corollary 2.6. $\alpha_7\geq 227^{1/3}\approx 6.1002$.

Proof. By Theorem 2.4, $r_p(\mathbb{F}_p^{m+k})\geq r_p(\mathbb{F}_p^m)\,r_p(\mathbb{F}_p^k)$, so $\log r_p(\mathbb{F}_p^n)$ is superadditive and by Fekete's lemma $\alpha_p=\lim_n r_p(\mathbb{F}_p^n)^{1/n}=\sup_n r_p(\mathbb{F}_p^n)^{1/n}$. Taking $n=3$ and Theorem 2.2 gives the claim. □

Remark 2.7. Theorem 2.6 is not the best known bound: [3, Thm. 1.1] gives $\alpha_7 \geq 7^{20/21} \approx 6.3805$ and $\alpha_{11} \geq 11^{54/55} \approx 10.5307$ (see the introduction), and no three-dimensional set can reach these values because $6.3805^3 > 242$ and $10.5307^3 > 1085$. For $p = 11$ the bound $1010^{1/3} \approx 10.0332$ does not even reach the older bound $\alpha_{11} \geq 11^{1/22}10^{21/22} \approx 10.0434$ of Frankl, Graham and Rödl ([1, Table 1]), for which a line-free set of at least 1014 points in $\mathbb{F}_{11}^3$ would be needed.

3 The two sets

3.1 Encoding and verification

A point $(x, y, z) \in \mathbb{F}_p^3$ is encoded by the integer $x + py + p^2z \in [0, p^3)$, with coordinates in $\{0, \ldots, p-1\}$. The data files `data/linefree_F7_227.txt` and `data/linefree_F11_1010.txt` list the encoded points (the `_xyz` files list coordinates); SHA-256 hashes are in `data/SHA256SUMS`:

```
linefree_F7_227.txt     fc505c9d673a4d390ba665a9333d35998ea1bfa34ab03e9b2c6cbad9d5623636
linefree_F11_1010.txt   3c65acbb753817d20bc5a7fc118cddef9d43ead14d0ee5be425febdc11070010
```

The sets are also drawn layer by layer in Sections A and B.

Proof of Theorems 2.2 and 2.3. $\mathbb{F}_p^3$ has $p^2(p^2+p+1)$ lines: 2793 for $p = 7$ and 16093 for $p = 11$. For each base point a and each nonzero direction d one checks that some $a + td$ lies outside the set. This finite verification was carried out by three independent programs: the Python script `data/verify.py` (all pairs (a, d), $d \neq 0$), a C implementation inside the search program, and the Lean kernel (Section 5); an independent script written by a second party also confirmed that no line is contained in either set. □

3.2 Statistics

Let n_j be the number of lines meeting the set in exactly j points. Then

$$\begin{aligned} S_7 &: \quad (n_0, \ldots, n_7) = (26, 34, 112, 282, 603, 993, 743, \mathbf{0}), \\ S_{11} &: \quad (n_0, \ldots, n_{11}) = (91, 99, 58, 19, 18, 103, 503, 1954, 4898, 5909, 2441, \mathbf{0}). \end{aligned}$$

(As a consistency check, $\sum_j jn_j = |S|(p^2+p+1)$, e.g. $12939 = 227 \cdot 57$.)

Both sets are maximal with respect to inclusion: adding any of the missing points creates a full line. Concerning structure:

- In the parallel class of planes $x+2y+3z = c$ of $\mathbb{F}_7^3$, the planes meet S_7 in $14, 34, 35, 36, 36, 36, 36$ points ($36 = (p-1)^2 = r_7(\mathbb{F}_7^2)$ is the maximum for a plane). The algebraic 225-point set of [1] consists of six layers of 36 points and one layer of 9 points. 105 points of S_7 have a "local line profile" (the multiset of $|L \cap S_7|$ over the 57 lines L through the point) shared by no other point of S_7; these points span $\mathbb{F}_7^3$ affinely, so the stabiliser of S_7 in $\mathrm{AGL}(3, 7)$ is trivial.

- In the parallel class $x + 7y + 7z = c$ of $\mathbb{F}_{11}^3$, ten planes contain $100 = (p-1)^2$ points of S_{11} and one plane contains 10, i.e. S_{11} has the coarse shape "ten maximal planes plus a sparse plane" of hypercube-type constructions, with 10 extra points instead of the $(p-1)/2 = 5$ of [1, Thm. 1.7].

4 How the sets were found

We searched for k-subsets of $\mathbb{F}_p^3$ minimising the number of fully contained lines.

1. *Tabu search.* A swap move removes a point of the current set and adds a point of the complement. Gains are maintained incrementally (for a point of the set, the number of full lines through it; for a point outside, the number of lines through it with $p-1$ points of the set); the best swap is chosen among all pairs, with random tie-breaking, short random tabu tenures and aspiration. Many *short* restarts work best: at $p=7$, $k=225$, restarts of 400–800 iterations gave about 40 solutions per 150 CPU-seconds, against 12 with 3000 and 10 with 6000 iterations; a two-stage move rule, adaptive line weights and greedy initialisation were all worse. With this method 224 points are found in about one restart in eight, 225 in one in two hundred, and a 226-point set was found after about five CPU-minutes; no 227-point set appeared in a further hour. For $p=11$, sets of size 1006–1008 were found in one to five minutes.

2. *Iterated local search.* Restart the same tabu search from a known solution with 10–30 randomly chosen points exchanged and the target size increased by one. This is much stronger: from a 225-point set, 226-point sets were found 61 times in 150 CPU-seconds (with 10 exchanged points), against once for random restarts. From the 226-point set, S_7 was found after about two CPU-minutes (20 exchanged points; S_7 shares 199 points with the 226-point set). For $p=11$, chaining this step from a 1008-point set gave 1009 and then S_{11}.

A SAT-based neighbourhood test (re-optimising exactly all points in one plane, or in two parallel planes, with the CaDiCaL solver) showed that the 226-point set cannot be enlarged by such local changes, which is consistent with the need for larger perturbations.

Negative results (not theorems). These are reported only to document the search effort.

- $p=7$: no 228-point set in about 85 CPU-minutes of iterated local search, including a random walk over 227-point solutions.
- $p=11$: no 1011-point set in about 47 CPU-minutes.
- $p=5$: 70-point sets are found within a few hundred iterations, but no 71-point set was found (several CPU-minutes of tabu search and iterated local search). We do not know whether $r_5(\mathbb{F}_5^3)=70$.
- $p=13$: sets of size 1731 were found within two minutes, but none of size 1735 (the bound 1734 of [1] plus one) in about 35 CPU-minutes of random restarts; iterated local search was not tried for $p=13$. Note that the cone construction gives 1740 points (see the introduction), so the search did not even reach the best known value for $p=13$.
- In $\mathbb{F}_5^4$ the search reached only $280 = 70\cdot 4$ points, so it gave no improvement of α_5.

5 Formal verification

Both theorems are proved in Lean 4.34.1 using only the core library and `Std` (no Mathlib), in the self-contained files `lean/Main.lean` ($p=7$) and `lean/Main11.lean` ($p=11$). No `sorry`, `axiom` or `native_decide` is used. The main statements and all definitions they depend on (for $p=7$; the $p=11$ file is identical with 11 and 1010) are:

```
abbrev Pt := Fin 7 × Fin 7 × Fin 7
def add (a b : Pt) : Pt := (a.1 + b.1, a.2.1 + b.2.1, a.2.2 + b.2.2)
def smul (t : Fin 7) (d : Pt) : Pt := (t * d.1, t * d.2.1, t * d.2.2)
def LineFree (L : List Pt) : Prop :=
  ∀ a d : Pt, d ≠ (0, 0, 0) → ∃ t : Fin 7, add a (smul t d) ∉ L
theorem main : ∃ L : List Pt, L.Nodup ∧ L.length = 227 ∧ LineFree L

def vadd (a b : List (Fin 7)) : List (Fin 7) := List.zipWith (· + ·) a b
def vsmul (t : Fin 7) (d : List (Fin 7)) : List (Fin 7) := d.map (t * ·)
def LineFreeN (n : Nat) (L : List (List (Fin 7))) : Prop :=
  ∀ a d : List (Fin 7), a.length = n → d.length = n → (∃ x ∈ d, x ≠ 0) →
    ∃ t : Fin 7, vadd a (vsmul t d) ∉ L
theorem main_pow (k : Nat) : ∃ M : List (List (Fin 7)), M.Nodup ∧ M.length = 227 ^ k ∧
    (∀ v ∈ M, v.length = 3 * k) ∧ LineFreeN (3 * k) M
```

Proof architecture. The witness `theSet` is the explicit list of points, sorted lexicographically. A bitmask with bit $p^2x + py + z$ set for each point is tied to it by the kernel-checked equation `theSet = allPts.filter inMask`, where `allPts` lists all p^3 points. `allPts` has no duplicates, because it maps onto `List.range` p^3 under the lexicographic index; this gives `theSet.Nodup`. The length is checked by `decide +kernel`. For line-freeness, a Boolean checker `checkAt a` is evaluated by `decide +kernel` on chunks of base points (7 slices for $p = 7$, 121 rows for $p = 11$); the chunks are glued by `match` on `Fin` and combined with soundness lemmas (`List.all_eq_true`, `List.any_eq_true`, membership of every point in `allPts`). For $p = 7$ the checker tries all 342 nonzero directions. For $p = 11$ it tries only directions whose first nonzero coordinate is 1. The reduction is proved in the file: a kernel check shows that $u^{-1}d$ is normalised for every $d \neq 0$, where u is the leading coordinate of d, and associativity of multiplication in `Fin 11` (checked by `decide +kernel`) gives $s(u^{-1}d) = (su^{-1})d$. `main_pow` is a structural proof: the product of two line-free, duplicate-free lists of vectors of fixed lengths is line-free and duplicate-free, of the product size (Theorem 2.4). It is proved by induction on k and instantiated with `main`'s witness.

Statement audit. (i) `Fin 7` addition and multiplication are taken modulo 7, so `Pt` with `add` and `smul` is the vector space $\mathbb{F}_7^3$. (ii) `LineFree L` says: for every base point a and every direction $d \neq 0$, some point $a + td$ is not in `L`, i.e. `L` contains no line. All pairs (a, d) are quantified; the direction normalisation for $p = 11$ happens only inside the proof. (iii) `L.Nodup` together with `L.length = 227` says that `L` consists of 227 distinct points. (iv) In `main_pow`, vectors of $\mathbb{F}_7^{3k}$ are lists of length $3k$, and this length is part of the statement; "$d \neq 0$" is `∃ x ∈ d, x ≠ 0`. For $k = 0$ the statement is the trivial $r_7(\mathbb{F}_7^0) \geq 1$. (v) Both theorems are existential statements whose witness is explicit, so there are no hypotheses that could be vacuous. A separate control file (`lean/NegTest.lean`, `#eval` only) checks that the same checker rejects the set with the line $\{(t, 0, 0)\}$ added, and the set with any single missing point added.

Axioms and resources. `#print axioms` reports, for both files,

```
'main' depends on axioms: [propext, Classical.choice, Quot.sound]
'main_pow' depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are the standard axioms; no `Lean.ofReduceBool` appears. Checking takes about 140 s with peak memory 2.2 GB for $p = 7$, and about 550 s with 0.96 GB for $p = 11$, on a shared machine. The outputs are in `lean/lean_output.txt` and `lean/lean_output11.txt`.

6 Limitations and open questions

- Novelty is claimed to our knowledge as of October 8, 2026, based on a literature search; unpublished improvements cannot be excluded.
- The exact values remain open: $227 \le r_7(\mathbb{F}_7^3) \le 242$ and $1010 \le r_{11}(\mathbb{F}_{11}^3) \le 1085$. The search gives no evidence that our sets are optimal.
- The bound $\alpha_7 \ge 227^{1/3}$ of Theorem 2.6 is weaker than the known bound $\alpha_7 \ge 7^{20/21}$ of [3]; for $p = 5, 7, 11$ no three-dimensional line-free set can improve the bounds for α_p that follow from [3].
- Not formalised: the existence of the limit α_7 (Fekete's lemma), and the comparison with the literature.
- The sets are unstructured. A structural explanation of why random search beats the algebraic constructions for $p = 7, 11$ might lead to improved general constructions of the form $(p-1)^3 + cp^\beta$. For $p = 11$, about ten points beyond the hypercube fit into one sparse plane.
- Is $r_5(\mathbb{F}_5^3) = 70$? This seems within reach of a careful exhaustive search with symmetry reduction, but a formal certificate would be substantially harder.

A The 227-point line-free set in $\mathbb{F}_7^3$

Layers $z = 0, \dots, 6$; in each layer the rows are $y = 0, \dots, 6$ (top to bottom) and the columns are $x = 0, \dots, 6$ (left to right). The symbol `#` marks a point of S_7; the layer sizes are given in parentheses.

```
z=0      z=1      z=2      z=3      z=4      z=5      z=6
..#.###  #.###.#  #..####  #.#.###  #..#.##  #.#.#..  #.##.#.
####.#.  ###.##.  .#.####  #..####  .######  #.#.#..  ##.##.#
####..#  ####.##  .##.###  #....#.  ###.##.  ###..##  ####.##
##.###.  ##.#...  ##.#.##  .######  ##.####  ....##.  #.##.##
.####.#  ####.#.  ..###.#  #..####  ##.####  ##..###  ..##.##
#.###.#  ##.###.  ..####.  ###.#.#  .#.#..#  ###....  #.#####
##..###  ..#.###  .###.#.  #.#####  #..##.#  .#..###  ###.##.
(34)     (33)     (32)     (34)     (34)     (25)     (35)
```

B The 1010-point line-free set in $\mathbb{F}_{11}^3$

Same conventions, with layers $z = 0, \dots, 10$, rows $y = 0, \dots, 10$ and columns $x = 0, \dots, 10$.

```
z=0          z=1          z=2          z=3          z=4          z=5
###.##.#.##  #.#######.#  #####.##...  #########.#  ##.###..#..  .##.##.####
#..#.######  ##.##.#####  ######.##.#  ##.####.###  ##.###.####  #########..
##.##.####.  ###.#####.#  ##.#.######  #.##...###.  ######..##.  ###.###.###
..######..#  ##.########  ##.###.####  ###.######.  ###..######  ##.##.#.###
##.######.#  #####..####  .#########.  #.#.###.###  ##.####.##.  .#..#######
#######..##  #######..#.  #.#.###.#.#  ##.####.##.  .##########  #.#..####.#
#####.####.  ######..###  ######..###  .#######.##  #########.#  ########.#.
###.###.##.  ####.##.#.#  .###..####.  ####...####  ##.####...#  ..######.##
##.####.###  .#.#####..#  ###..######  #.##.###.#.  #.#.#.#####  .####..####
```

```
.#########.  .###.####..  #.######.##  #..#.#.####  #.#.#.####.  ####..##..#
####.##..##  ########..#  #.#.######.  .###....##.  #.###..##.#  ##.########
(92)         (94)         (92)         (86)         (89)         (91)

z=6          z=7          z=8          z=9          z=10
#.##..####.  ##.#######.  ###.#######  #######.###  ####..#####
###.######.  ..##.######  ..#########  ##...######  ###.####.#.
####.##..##  .########.#  ####.######  .##########  ...########
######.####  ##...##.###  ########.##  #.#..##.#..  #####.##.#.
##.#.######  #####.##.##  #..###.#.##  #####.#.###  ######.##.#
########.##  #.#.#######  #####.##.##  #.###.#.#.#  ##.###.####
..#.#######  #.###..####  ##.#.##....  #...##.####  .###.#..###
#####.###.#  #####.#.#..  .#.########  #.####.####  ##.#######.
#.##.##.#.#  ##.#####.##  ######.##.#  #..#.#####.  .##..##.###
#####.#####  #.####.####  #######.##.  ###.#.##.##  #######.###
###.##.####  ########.#.  ###.#.#.###  ######..###  .######.###
(96)         (93)         (95)         (89)         (93)
```

References

[1] C. Elsholtz, J. Führer, E. Füredi, B. Kovács, P. P. Pach, D. G. Simon, N. Velich, *Maximal line-free sets in* $\mathbb{F}_p^n$, Period. Math. Hungar. **90** (2025), 7–21, https://doi.org/10.1007/s10998-024-00617-x; arXiv:2310.03382 (v2, May 2024). https://arxiv.org/abs/2310.03382

[2] B. Kovács, *A superlinear improvement on line-free sets in* $\mathbb{F}_p^3$, arXiv:2605.23437 (May 2026, revised September 2026). https://arxiv.org/abs/2605.23437

[3] J. Führer, V. Taranchuk, *Large line-free sets and their applications*, arXiv:2403.18611 (v1 March 2024, v2 February 2026). https://arxiv.org/abs/2403.18611

[4] A. Bishnoi (question), D. Zare (answer), *Blocking sets in three dimensional finite affine spaces*, MathOverflow, January 2016. https://mathoverflow.net/q/229442

[5] S. Ball, A. Blokhuis, *On the size of a double blocking set in* $PG(2,q)$, Finite Fields Appl. **2** (1996), 125–137.

[6] M. Braun, A. Kohnert, A. Wassermann, *Construction of* (n,r)*-arcs in* $PG(2,q)$, Innov. Incidence Geom. **1** (2005), 133–141.

[7] B. Csajbók, T. Héger, *Double blocking sets of size* $3q-1$ *in* $PG(2,q)$, European J. Combin. **78** (2019), 73–89. arXiv:1805.01267.

[8] A. Bishnoi, J. D'haeseleer, D. Gijswijt, A. Potukuchi, *Blocking sets, minimal codes and trifferent codes*, arXiv:2301.09457 (2023).

[9] R. E. Jamison, *Covering finite fields with cosets of subspaces*, J. Combin. Theory Ser. A **22** (1977), 253–266.

[10] A. E. Brouwer, A. Schrijver, *The blocking number of an affine space*, J. Combin. Theory Ser. A **24** (1978), 251–253.